🏥 Healthcare Operations Analytics

Project Overview

This project analyzes historical patient encounter data to uncover operational inefficiencies and provide actionable insights that can help improve patient flow, reduce wait times, and enhance overall patient experience.

The primary objective of this analysis is to identify patterns in patient demand, operational bottlenecks, and factors affecting patient experience, and provide data-driven recommendations to optimize hospital operations**.**

Dataset Summary

The dataset consists of **9,000+ patient encounter records** collected between **April 2023 and October 2024**.

Key Features:

- **Patient Demographic**: Age, Gender, Race
- **Visit Details**: Admission Date & Time, Visit Type (Emergency, Walk-in, Scheduled)
- **Operational Metrics**: Department Referral, Wait Time
- **Outcome Variables**: Admission Flag (Admitted / Not Admitted)
- **Experience Metrics**: Patient Satisfaction Score
- **Relational Data**: Doctor ID linked to doctor details

Exploratory Data Analysis

Initial exploration was conducted to understand the structure and quality of the dataset.

Key Steps:

- Validated total number of records
- Reviewed sample data for structure and consistency
- Identified missing values across columns

- Checked date range of patient encounters
- Analyzed basic distributions (department, gender, etc.)

Key Findings:

- Approximately **13% missing values** in satisfaction scores likely represent **non-response rather than data errors**
- Null values in department and doctor_id indicating **data quality limitations**
- Date column initially stored as **VARCHAR**, requiring transformation
- Presence of inconsistent entries in gender values
- Patient data spans over **15 months**, enabling time-based trend analysis
- Uneven distribution of patients across departments

Data Cleaning

Data Cleaning Actions:

- Converted date column to proper date format
- Replaced null department values with “Unknown”
- Standardized invalid gender entries
- Retained nulls in satisfaction score and doctor_id to preserve data integrity

Data Analysis using SQL (Business Transactions)

The analysis focused on answering key business questions related to hospital operations.

1. Patient Volume Trends:

Patient volume was analyzed over time (monthly and yearly) to identify growth patterns and seasonality.

Insight:

Patient demand is **stable at a monthly level but growing steadily year-over-year**.

The hospital is **not experiencing sudden surges.** But there is a **clear upward trend in overall demand.**

2. Visit Type Distribution

Patient visits were segmented into Emergency, Walk-in, and Scheduled.

Insight:

Patient demand is **balanced across Emergency, Walk-in, and Scheduled visits**, with a slight skew toward Emergency cases.

Hospital should prioritize efficient handling of **Emergency cases**, as they are the largest and most unpredictable segment.

3. Wait Time Analysis

Average wait times were calculated across the hospital and by department.

Insight:

Wait times are **consistently high across all departments**, with only minor variation between them.The issue is **system-wide**, not isolated to one department.

4. Service Efficiency

The percentage of patients waiting more than 30 minutes was calculated.

Insight:
A considerable portion of patients experience delays beyond acceptable thresholds, highlighting **service inefficiency**.

5. Peak Hour Analysis

Patient arrivals were analyzed by hour of the day.

Insight: Patient arrivals are **fairly evenly spread throughout the day**, with mild peaks rather than sharp spikes.

6. Department-Hour Bottlenecks

Wait times were analyzed across department and hour combinations.

Insight:
Operational strain is driven by **specific department-time intersections**, not just overall volume.

7. Patient Satisfaction Analysis

Satisfaction scores were analyzed across departments.

Insight:
Patient experience is similar across departments and is relatively on the lower side.

8. Admission Rate Analysis

Admission rates were calculated by department.

Insight:
Admission rates are very similar across all departments (~48%–51%)

9. Wait Time vs Admission Relationship

Wait times were compared with admission likelihood.

Insight:
Longer wait times are not directly associated with higher admissions, although there is a correlation between the two.

10. Shift-Based Operational Analysis

Patient load and wait times were analyzed across doctor shifts.

Insight:
Operational efficiency varies across shifts, indicating **misalignment in staffing and demand**. Night shift doctors are handling more patients.

Business Recommendations

- **Reduce wait times below 30 minutes:** Optimize triage and intake processes, as most patients currently experience delays that directly impact satisfaction.
- **Fix department-hour bottlenecks:** Target specific high-delay combinations (e.g., Neurology nights, Physiotherapy mornings) instead of broad system changes.
- **Rebalance staffing and workload:** Align doctor shifts and patient assignments with actual demand to avoid overloading certain shifts or doctors.
- **Improve efficiency before adding capacity:** Focus on process optimization and workflow improvements, since delays are driven by inefficiencies, not demand spikes.